Lambda 1 | Cantera vs NN NRMSE | 10,000 updates → long continuation

Repaired 10 ms rollout; cell top: short, bottom: long. Medians use the SAME initial conditions within each model row.

Model	Common ICs	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	Complete rollouts
ANN10	17	0.204 → 0.172	0.799 → 0.683	4.12 → 3.8	0.99 → 0.963	0.613 → 0.524	2.46 → 1.81	0.265 → 0.203	2.84 → 6.25	2.51 → 27.4	0 → 0	17/18 → 18/18
P-only DeepONet10	18	0.0793 → 0.145	0.461 → 0.168	0.908 → 1.36	5.52 → 0.746	0.575 → 0.523	0.97 → 0.554	0.102 → 0.087	6.43 → 1.25	2.05 → 1.05	0 → 0	18/18 → 18/18
ANN10 + P-only10	18	0.0437 → 0.0707	0.454 → 0.176	0.69 → 0.659	5.28 → 0.775	0.594 → 0.436	0.985 → 0.735	0.0777 → 0.0772	2.95 → 1.13	1.44 → 1.03	0 → 0	18/18 → 18/18
T/P DeepONet10	8	0.293 → 0.34	0.749 → 0.946	3.78 → 1.85	1.09 → 1.48	0.544 → 0.578	1.11 → 1.04	0.496 → 0.676	7.02 → 47.9	22.7 → 25.7	0 → 0	9/18 → 14/18
ANN10 + T/P10	18	0.0803 → 0.0902	0.41 → 0.189	0.932 → 0.888	0.819 → 0.588	0.535 → 0.4	0.822 → 0.518	0.103 → 0.0959	0.938 → 0.905	1.16 → 0.992	0 → 0	18/18 → 18/18
T/P/phi DeepONet10	8	0.697 → 0.201	0.966 → 0.956	21.1 → 11.7	70.9 → 2.53	0.661 → 0.458	7.13 → 3.07	0.72 → 0.646	8.02 → 35.6	1 → 5.64	0 → 0	11/18 → 13/18
ANN10 + T/P/phi10	17	0.395 → 0.209	0.881 → 0.347	2.89 → 1.2	3.51 → 1.07	0.521 → 0.549	8.19 → 1.15	0.507 → 0.232	1.22 → 1.29	1.15 → 1.27	0 → 0	17/18 → 18/18

NRMSE = RMSE(Cantera, NN) / RMS(Cantera). Exact user util.py definition; dimensionless ratio, without ×100.

Common ICs are the intersection of baseline10k, soft10k and soft-long for each row. Row cohorts differ; use this table for within-model changes.

Long models ended at the time limit (joint plateau 0/120); actual endpoints: lambda1 41k–81k, lambda100 35k–84k, lambda1000 29k–62k.

Lambda 100 | Cantera vs NN NRMSE | 10,000 updates → long continuation

Repaired 10 ms rollout; cell top: short, bottom: long. Medians use the SAME initial conditions within each model row.

Model	Common ICs	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	Complete rollouts
ANN10	18	0.658 → 0.656	0.88 → 0.911	1 → 1	1 → 1	1.3 → 0.955	22.7 → 32.3	1 → 1	1.06 → 13.3	84.2 → 5.98e+03	0 → 0	18/18 → 18/18
P-only DeepONet10	1	0.593 → 0.602	0.718 → 0.66	1 → 1	1 → 1	9.71 → 7.39	28.3 → 30.2	1 → 1	1.02 → 0.997	3.14 → 1.02	0 → 0	6/18 → 1/18
ANN10 + P-only10	13	0.641 → 0.416	1.04 → 0.919	1 → 1	1 → 1.44	2.72 → 1.44	26.3 → 19.8	1 → 1	1.17 → 8.33	43.5 → 1.52e+03	0 → 0	18/18 → 13/18
T/P DeepONet10	2	0.179 → 0.125	0.558 → 0.645	8.19e+07 → 4.41e+04	2.02e+08 → 2.22	0.633 → 0.739	3.58e+10 → 4.12e+10	1.24e+05 → 1.69e+05	1 → 4.15	1 → 1	0 → 0	18/18 → 3/18
ANN10 + T/P10	18	0.332 → 0.204	0.973 → 0.783	1.42 → 1	11.5 → 3.18	2.46 → 0.925	1 → 2.98	0.561 → 0.442	324 → 303	2.03e+03 → 1.32e+04	0 → 0	18/18 → 18/18
T/P/phi DeepONet10	4	0.239 → 0.121	0.59 → 0.191	1.04e+08 → 3.06	3.20e+08 → 1.24e+03	0.812 → 0.483	1.17e+10 → 2.24e+07	1.55e+05 → 0.986	1 → 13.7	1 → 5.37e+04	0 → 0	16/18 → 4/18
ANN10 + T/P/phi10	18	0.348 → 0.669	0.995 → 0.845	1.78 → 2.62	8.95 → 3.78	2.93 → 0.97	1 → 18.8	0.583 → 0.999	280 → 205	2.01e+03 → 2.12e+04	0 → 0	18/18 → 18/18

NRMSE = RMSE(Cantera, NN) / RMS(Cantera). Exact user util.py definition; dimensionless ratio, without ×100.

Common ICs are the intersection of baseline10k, soft10k and soft-long for each row. Row cohorts differ; use this table for within-model changes.

Long models ended at the time limit (joint plateau 0/120); actual endpoints: lambda1 41k–81k, lambda100 35k–84k, lambda1000 29k–62k.

Lambda 1000 | Cantera vs NN NRMSE | 10,000 updates → long continuation

Repaired 10 ms rollout; cell top: short, bottom: long. Medians use the SAME initial conditions within each model row.

Model	Common ICs	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	Complete rollouts
ANN10	18	0.67 → 0.635	1.55 → 1.21	1 → 1	66.6 → 1	4.22 → 2.85	4.11 → 24.7	1 → 1	0.999 → 1	704 → 14.7	0 → 0	18/18 → 18/18
P-only DeepONet10	18	0.555 → 0.553	1.34 → 1.37	1 → 1	1 → 1	2.77 → 3.14	1 → 1	1 → 1	1.35e+04 → 1.24e+04	1 → 1	0 → 0	18/18 → 18/18
ANN10 + P-only10	18	0.542 → 0.58	1.64 → 1.46	1 → 1	1 → 0.999	5 → 4.07	1 → 11.9	1 → 1	0.998 → 0.999	815 → 17.2	0 → 0	18/18 → 18/18
T/P DeepONet10	17	0.553 → 0.545	1.56 → 1.8	1 → 1	1 → 1	4.96 → 5.09	1 → 1	1 → 1	627 → 0.983	1 → 1.81	0 → 0	18/18 → 18/18
ANN10 + T/P10	18	0.542 → 0.544	1.64 → 1.64	1 → 1	1 → 1	5 → 5.01	1 → 1	1 → 1	0.998 → 0.999	761 → 8.34	0 → 0	18/18 → 18/18
T/P/phi DeepONet10	17	0.533 → 0.542	1.47 → 1.48	1 → 1	1 → 1	4.94 → 4.94	1 → 1	1 → 1	4.69 → 0.987	1 → 2.91	0 → 0	18/18 → 18/18
ANN10 + T/P/phi10	18	0.544 → 0.543	1.64 → 1.64	1 → 1	1 → 0.999	4.99 → 5.01	1 → 1	1 → 1	0.997 → 1	830 → 3.85	0 → 0	18/18 → 18/18

NRMSE = RMSE(Cantera, NN) / RMS(Cantera). Exact user util.py definition; dimensionless ratio, without ×100.

Common ICs are the intersection of baseline10k, soft10k and soft-long for each row. Row cohorts differ; use this table for within-model changes.

Long models ended at the time limit (joint plateau 0/120); actual endpoints: lambda1 41k–81k, lambda100 35k–84k, lambda1000 29k–62k.